<u>U2-D6 Homework</u> – Difference of Squares (and Cubes) **Name:**

On this assignment, squares and cubes are in separate sections. In the future, you need to be able to recognize which pattern applies to a given problem.

Factor each difference of squares.

1. $x^2 - 169$

2. $4x^2 - z^2$

3. $x^8 - 16$

4. $9y^6 - x^{14}$

5. $3c^2 - 75g^2$

6. $x^{13} - 36x^3$

Factor each sum or difference of cubes. Simplify each term of your answer.

First, write out the formula:

7. $x^3 + 64$

8. $8x^3 - y^3$

9. $x^{21} + y^{15}$

10. $54x - 2z^3$

8. Find an equation in slope-intercept form for a line through the points $(4, -3)$ and $(-2,0)$.

10. Evaluate. $f(x) = \begin{cases} -x^2 - 3x & x < 4 \\ -2|x-1| + 9 & x \geq 4 \end{cases}$

a) $f(-6)$

b) $f(4)$

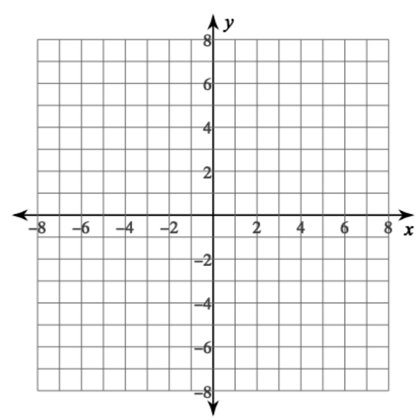
Solve by factoring.

11. $x^2 - 13x = -42$

12. $12x^2 + 8x - 15 = 0$

13. Graph.

$f(x) = \begin{cases} -2|x+3| + 5 & x < 1 \\ 4 & x \geq 1 \end{cases}$



Simplify completely. [That always implies no negative exponents left in answers and that an expression like 3^5 simplifies to 243. Also, **fractions, not decimals!**]

14. $\frac{12^9 b^9}{12^7 b^{13}}$

15. $3x^{-8}(3x^{-2})^{-4}$